AI for Good Sandbox Hackathon – Xtreme Vision Computer – TechTutor Governance Lab

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Github link: https://github.com/omarbatainaomar-blip/techtutor-governance-lab (private; shared with the reviewers CrashingGuru and jiayingmeng)

1. Introduction

Generative AI is entering UAE classrooms quickly: AI is now taught across public schools from the 2025–26 school year, and students already use chatbots every day. Two problems follow. First, general chatbots answer from model memory, give no source and answer confidently outside their scope, which is risky for minors' learning and for assessment. Second, the rules that decide whether an AI tutor may be used in a school are spread across several authorities (Ministry of Education, ADEK, UAE Data Office, AI Office) and documents with very different legal force. Approval is a manual document review, and nobody rehearses in advance what should happen when the tool goes wrong.

TechTutor Governance Lab addresses both problems in three layers. (1) **Knowledge bases:** a bilingual tutor KB (374 passages, 7 computing and AI topics) and a UAE policy KB built with the ITU reference ingestion code (1,863 passages from 28 official documents, each passage tagged with page, article number, binding scope and licence). (2) **Agents:** a grounded tutor plus five governance agents that act as the institutions and decide, under pre-agreed rules, whether the tutor may continue. (3) **Web interface:** a chat for students and a Governance Lab to run, replay and download each simulation. Everything runs locally (Ollama + Chroma), so no student data leaves the school machine. The pipeline is mapped to ITU-T Y.3172 and the ITU AI Readiness Framework 2.0, so the tutor is not a standalone app but a deployment tested against UAE policy before it reaches students.

2. Description of the use case

Users: secondary-school students (11–17) learning computing and AI in Arabic or English, their teachers and guardians, and the authorities that approve and monitor EdTech. **Gaps in existing solutions:** answers without evidence; weak Arabic support; governance done on paper with no record of why a decision was taken; and strategies, charters and laws cited as if they carried the same legal force. **How our solution closes them:**

- **Grounded tutor.** The tutor answers only from retrieved passages. After the model writes, every sentence is compared by meaning with the passages (bge-m3, works across Arabic and English); the system attaches the citation [n] itself and removes wrong ones. If less than half of the answer is supported, the student sees “not found” instead of an answer from memory. Prompt-injection and harmful requests are blocked before the model; uploaded files are treated as data only. In an evaluation across the 7 topics, 26 of 29 questions behaved as expected before the last fix, including 8 of 8 control cases (questions outside the KB such as BGP or Kubernetes were refused,

injection was blocked); answers grounded in the sources scored 0.55–1.00 support and answers from model memory 0–0.44.

- **Governance sandbox.** Five agents: TechTutor (solution provider), Policy & Legal Advisor, Ministry of Education, Data Protection Regulator and AI Governance Authority; the last three can suspend. Each agent has a mandate, searches only its own part of the KB and receives private information that the scenario releases stage by stage. Agents exchange QUESTION, ANSWER and NOTIFY messages.
- **Evidence gate.** Every quote must appear verbatim in a retrieved passage (Arabic-aware); quotes that drop a negation or change a number are rejected; relevance is scored; numbers that do not appear in the case or the sources are flagged as possible hallucinations and stay flagged when another agent repeats them.
- **Pre-agreed rules decide; the model only proposes.** R1: an authority BLOCKs with a verified, relevant quote of 8+ words from a **binding** UAE text → suspend. R1b: an authority rates the risk high with the same kind of quote → precautionary suspension. R2: information missing, failure or unsupported verdict → human review. R3: conditions → continue with conditions. R4: all approve with evidence → continue. R5: a suspension holds until every suspending authority approves again with evidence. Binding texts: PDPL 45/2021, Child Digital Safety 26/2025, Consumer Protection 15/2020 and the ADEK School Digital Policy; strategies, charters and guides only support conditions; DIFC/ADGM rules and international texts are comparisons.
- **Accountability.** Every release, response, message and decision goes into a SHA-256 hash-chained log that can be verified independently; each run produces a readable report and scores its own expected behaviour.

3. Use case requirement

SRC – Student questions and files (Arabic/English) and official policy documents. Required: guardian consent recorded before use; students warned not to paste identifiers (in S1 an audit found Emirates ID numbers in stored chats); every document kept with issuer, year and official URL.

C – School-hosted HTTP API (rate limits, session expiry, same-origin checks) and a document collector. Required: student data stays in the UAE unless PDPL Art. 22–23 conditions are met; no redirection of students to unapproved public chatbots (S3).

PP – ITU chunking (256 words, 30 overlap), page and article tagging, Arabic PDF repair, binding-scope and licence tags, bge-m3 embeddings. Required: retention limited to the purpose; identifiers masked before storage (proposed).

M – Local qwen2.5:7b via Ollama; an optional cloud model sits behind the same interface. Required: no training on student chats; model, code and KB versions hashed in every report; a tested fallback plan (AI Ethics Guidelines 3.5.2.2).

P – Sentence-level attribution and refusal in the tutor; evidence gate and rules R0–R5/R1b in the sandbox. Required: no profiling of students for marketing; every suspension or escalation reviewed by a human.

D – Decisions, conditions, rule text and verified quotes sent to the operator and authorities. Required: traceability of significant decisions (AI Ethics Guidelines 3.3.2.1).

SINK – Students (answers with sources), teachers and guardians, authorities and auditors (reports, verifiable logs). Required: guardians informed of data use; logs verifiable by a third party.

4. Mapping with ITU-T Y.3172 Readiness 2.0 Report, and domain-specific policies

Node	Use case (TechTutor, education)	Readiness 2.0 dimension	Documents in the KB (binding = can justify suspension)
SRC	Student questions and files; official policy documents with provenance	Data/Model Marketplace; Contextualization and Regional Impact	PDPL 45/2021, consent and data-subject rights (binding); Child Digital Safety 26/2025 (binding); UNESCO GenAI guidance, age-13 threshold (benchmark)
C	School-hosted API; document collector	Digital Infrastructure	PDPL Art. 22–23, cross-border transfer (binding); ADEK School Digital Policy (binding, emirate); DIFC Law 5/2020 (comparison)
PP	ITU chunking, Arabic repair, article and binding-scope tags, bge-m3	Cross-Domain Correlation Analysis (Arabic–English retrieval)	ITU-T Y.3172; ITU AI Readiness Report 2.0, REQ-024 uniform record format
M	Local qwen2.5:7b; optional cloud model	Collaboration with AI	UAE AI Charter 2024; AI Ethics Guidelines 2022, robustness and fallback (guidance); EU AI Act (benchmark)
P	Attribution check, evidence gate, rules R0–R5 and R1b	AI & Policies; Strategy Alignment	PDPL; Consumer Protection 15/2020, misleading advertising (binding); Child Digital Safety; National AI Strategy 2031 (guidance); UNICEF Guidance on AI and Children 3.0 (benchmark)
D	Decisions with conditions, rule text and quotes; human-review flag	Human Interface; Impacts of Humans in AI Integration	AI Ethics Guidelines 3.3.2.1, traceability (guidance); OECD AI Principles (benchmark)
SINK	Students, teachers, guardians, authorities, auditors	Level of Integration of AI in Workflows; AI for Inclusion (Arabic-first)	ADEK School Digital Policy (binding); Generative AI Guide 2023 (guidance)
MLFO / Sandbox	Coordinator and four staged scenarios with expected-behaviour checks	AI & Policies; Granular Priorities	ITU AI4G Sandbox summary; Y.3172 sandbox concept

Y.3172 targets machine learning in future networks; we use its pipeline and sandbox as an architectural analogy for an education AI service. Article numbers are given only where verified in the KB text.

5. Error scenarios.

Each scenario releases information in stages and only to selected agents, so no agent sees the whole picture. The corrective action is never improvised: it follows the rules in section 2. Final runs of 6 November 2026 (local CPU, 32–48 minutes each; every audit chain verified):

Scenario	What goes wrong (by stage)	Decisions	Key evidence
S1 Student data, minors, offshore model	1. Pilot keeps chats 12 months. 2. Audit finds Emirates ID numbers in chats; vendor plans an offshore cloud model. 3. A sibling opens another student's saved chats. 4. No guardian consent collected.	Escalate → Continue (wrong) → Suspend R1b → Suspend	29 verified / 0 rejected quotes
S2 Exam integrity	1. Citation drift known only to the tutor. 2. Teachers report verbatim submissions. 3. A wrong subnetting answer used by 300 students before a national assessment.	Escalate ×3 (R2)	No binding UAE text on AI content accuracy → gap G1
S3 Outage and shortcut	1. Inference server and policy checks down. 2. Operator proposes sending students to a free public chatbot and disabling checks. 3. Late news of a new ministry manual.	Escalate → Suspend R1b → Suspend	Data Regulator, PDPL; 28 / 1

Scenario	What goes wrong (by stage)	Decisions	Key evidence
S4 Ads from student chats (template scenario)	1. Paid “exam booster” pop-ups after wrong answers; children spend up to AED 600 a month. 2. Students labelled “struggling” from their chats; label sent to an ad network outside the UAE. 3. Operator cites one checkbox; a guardian complains (12-year-old with a learning difficulty).	Escalate → Suspend R1 (explicit BLOCK) → R5 holds	Child Digital Safety 26/2025: “Applying Child privacy and personal data protection features according to age group”

Judges' added scenario. S4 implements the template's example: the case escalates from a consumer-protection issue (stage 1, human review) to a children's data-privacy and ethics breach (stage 2, suspension by the Data Protection Regulator on a binding law), and the gap analysis recommends prohibiting profiling of children for marketing and enforcing consent and purpose limitation. A new scenario is a short YAML file (stages, who receives what, expected behaviour); on the day we can run its first stage live (about 10 minutes on CPU) and show complete runs as replays.

Where the solution faces controversy, and how it responds:

- **A wrong agent verdict (S1, stage 2).** The Data Regulator approved although Emirates IDs were about to go offshore. The scenario's expected-behaviour check marks this stage as failed in the report, and the next stage suspended the pilot. We kept the run unedited: the checks, not the model, are the safeguard.
- **An over-cautious small model.** Agents often wrote “this is a violation of the PDPL”, rated the risk high and still answered NEED_INFO. We added the precautionary rule R1b. Trade-off: it relies on the model's own risk rating and can suspend on a weak quote (S1 cites an ADEK clause on staff accounts), so every suspension is flagged for human review.
- **Law versus guidance (S2).** The tutor asked to suspend itself citing only guidance documents; the rules refused and escalated to a human. One can argue a national strategy should carry more weight; we chose legal force and report the missing binding text as a gap.
- **Run-to-run variation.** S3 was suspended in one run and only escalated in another, and 3 quotes were rejected for changed meaning. This is why decisions come from fixed rules, every step is logged, and humans review.

6. Policy gaps and standards gaps derived from the mapping exercise

Each run ends with a gap analysis that compares UAE passages with benchmarks; an absence check flags claims of “no UAE text” when a close UAE passage exists. We reviewed all 12 gaps from the final runs by hand and consolidate them below. One claim was wrong: agents twice called cross-border transfer “missing”, but PDPL Art. 22–23 cover it, which shows why gap claims need human review.

- **G1 Accuracy and assessment use of AI tutors (S2; node P/D; AI & Policies).** No binding UAE text; only guidance on traceability and fallback (AI Ethics Guidelines 3.3.2.1, 3.5.2.2). The EU AI Act treats AI that evaluates learning outcomes as high-risk. **Recommendation:** an MoE/ADEK standard for AI tutors: cited sources, accuracy monitoring, correction notices to affected students, and rules for use in graded work.
- **G2 Profiling and commercial targeting of children in EdTech (S4; node P; Granular Priorities).** Partial: Consumer Protection 15/2020 covers misleading

advertising, Child Digital Safety 26/2025 requires age-appropriate privacy features and PDPL requires consent, but no text explicitly bans profiling students for marketing. UNICEF: states should prohibit profiling or targeting children for commercial purposes. **Recommendation:** an explicit ban with sanctions in school AI procurement.

- **G3 Age threshold and guardian consent for generative AI in school (S1, S4; node SRC).** Partial: consent rules are general. UNESCO recommends a minimum age of 13 for independent use. **Recommendation:** an age threshold and verifiable guardian consent for AI tutors.
- **G4 Change management and continuity (S1, S3; node M/C).** Guidance only (fallback plan); nothing requires re-approval when the model or hosting changes, as in the offshore switch (S1) or the public-chatbot shortcut (S3). The EU AI Act requires a new conformity assessment after substantial changes. **Recommendation:** re-approval triggers for education AI.
- **G5 Retention of students' chat data (S1; node PP).** PDPL sets general purpose limitation but no maximum period for minors' chat logs. **Recommendation:** a sector retention limit and masking of identifiers.

Standards gaps. (a) There is no common record format for AI-policy knowledge bases; the ITU Readiness report asks for a uniform record format (REQ-024), and our manifest/JSONL schema (passage, page, article, issuer, binding scope, licence) is a candidate. (b) No standard marks the legal force of a policy text in machine-readable form, yet an AI governance agent cannot weigh evidence without it. (c) Y.3172 has no profile for policy nodes in education; we propose a sandbox profile with staged scenarios, pre-agreed rules and a verifiable audit log.

Limitations. Small local model (about 2 minutes per agent call; Arabic output sometimes awkward); the attribution check confirms meaning, not every detail, and does not check code; agents simulate institutional roles and do not represent their official positions; 14 listed documents could not be downloaded automatically. The tutor KB consists of study notes written with AI assistance, disclosed in the repository.

References (not counted in the page limit)

[1] Federal Decree-Law No. 45 of 2021 on the Protection of Personal Data (PDPL). uaelegislation.gov.ae/en/legislations/1972

[2] Federal Decree-Law No. 26 of 2025 Regarding Child Digital Safety. uaelegislation.gov.ae/en/legislations/3912

[3] Federal Law No. 15 of 2020 on Consumer Protection. uaelegislation.gov.ae/en/legislations/1454 (copy from the organizers' UAE dataset)

[4] ADEK School Digital Policy v1.1 (2024). adek.gov.ae

[5] UAE AI Office: AI Ethics Principles & Guidelines (2022); UAE Charter for the Development & Use of AI (2024); National Strategy for AI 2031; Generative AI Guide (2023). ai.gov.ae, u.ae

[6] ITU-T Y.3172 (2019), Architectural framework for machine learning in future networks including IMT-2020. itu.int/rec/T-REC-Y.3172

[7] ITU AI Ready Analysis: Towards a Standardized Readiness Framework, Report 2.0, and reference code (MIT). github.com/CrashingGuru/ITUAIReadiness

[8] Regulation (EU) 2024/1689 (AI Act); UNESCO Guidance for Generative AI in Education and Research (2023); UNICEF Guidance on AI and Children 3.0 (2025); OECD AI Principles (2024).